



PASSENGER DELAY PREDICTION AND FORECASTING INSIGHTS

CHARLOTTE DOUGLAS INTERNATIONAL AIRPORT (CLT)

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EXECUTIVE SUMMARY

Charlotte Douglas International Airport (CLT) faces growing challenges tied to passenger growth, operational bottlenecks, and weather-related disruptions. This report introduces two critical data-driven tools: one predicts whether a flight will be delayed, and the other forecasts expected passenger traffic will be expected through mid-2026. Improved decision-making to improve passenger experience, to optimize operations and crucially protect revenue can be achieved via these tools. Ultimately, CLT stands to benefit substantially by integrating these analytics capabilities into its strategic planning framework. Doing so would position the airport to deliver a more seamless passenger experience, increase revenue resilience, and strengthen its competitive standing among national aviation hubs.

PREDICTING FLIGHT DELAYS: A PROACTIVE APPROACH

Charlotte Douglas International Airport (CLT), as a major hub with over 45 million annual passengers and American Airlines' second-largest operation, experiences elevated exposure to delay propagation. A predictive model enables pre-emptive planning across one of the nation's most congested terminal networks.

We developed a flight delay prediction tool that estimates the likelihood of a departure being delayed by 15 minutes or more. The system uses historical flight and weather data to identify conditions most likely to disrupt operations and departure times for passengers.



Each delayed flight contributes to gate congestion, staffing complications, and lost retail revenue. Through the CLT Appendix file, we estimated a loss of \$15 per passenger due to severe delays in 2023. Using these predictive insights, we can optimize gate assignments, reduce disruptions, and proactively communicate with passengers.

We experimented with multiple classification models including decision trees, XGBoost and Neural Networks. After careful performance evaluation, the Random Forest classifier emerged as the best fit due to its robustness to noise, ability to handle categorical and numerical variables, and resilience against overfitting. Input data was cleaned, engineered, and balanced to address class imbalance from low delay incidence.

RANDOM FOREST CLASSIFIER

model was determined to be appropriate here. It achieved approximately **80%** accuracy with a sharp focus on precision (as planned) and demonstrated strong performance in identifying actual delays with minimal false positives. The delay risk is most closely associated with departure time, aircraft type, weather conditions, and route distance. Decision Trees performed well on recall but lacked granularity in identifying complex delay patterns. Gradient Boosting slightly outperformed in accuracy but required longer training time and was more difficult to interpret for operational use. Overall, Random Forest offered the most practical deployment value.



COMPARISON among the three models tested,

random forest achieved the best overall balance, with an accuracy of **68.9%** and the highest precision at **44.5%**, making it more reliable in identifying true delays with fewer false alarms. While the decision tree yielded slightly higher recall (**62.0%**), it did so at the cost of lower precision and interpretability. XGBoost delivered the best recall overall (**71.5%**) but had the lowest accuracy (**60.3%**). In operational contexts where real-time interpretability, scalability, and false positive reduction are critical, random forest offered the most practical compromise between predictive power and deployment efficiency.

FORECASTING PASSENGER GROWTH: PLANNING AHEAD

We also developed a model to forecast monthly passenger volumes through May 2026 using T-100 airline data. This model captures seasonal peaks and overall passenger growth trends.

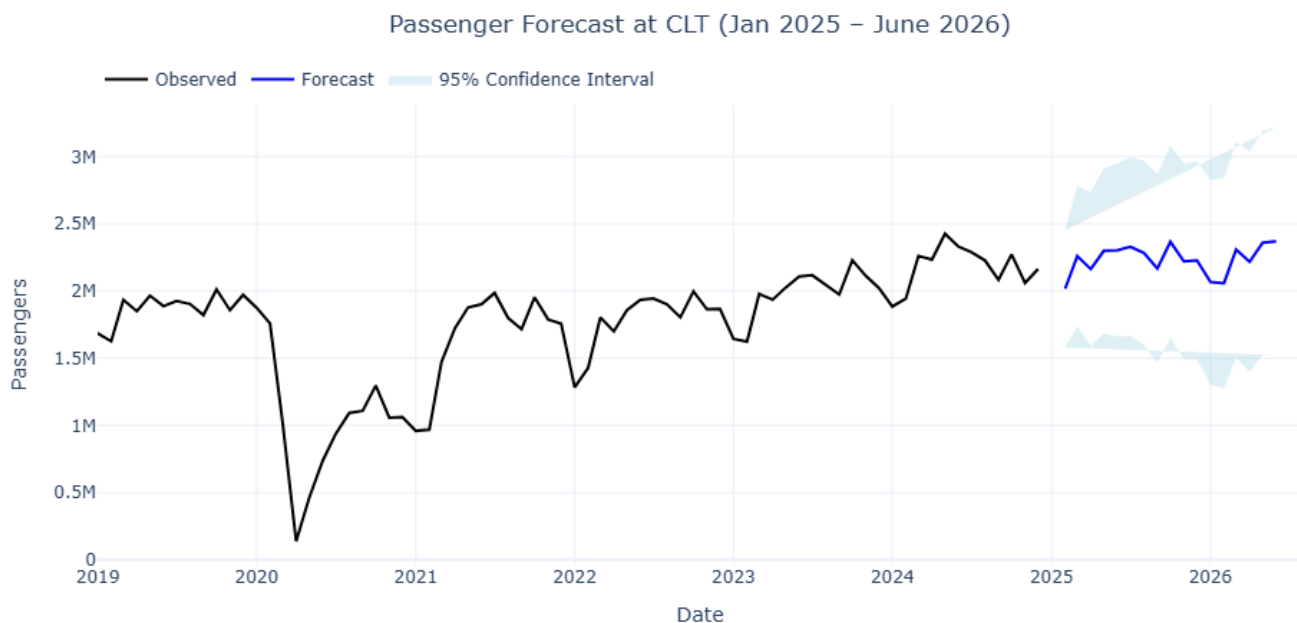
Passenger traffic directly drives revenue and resource demand. As per the CLT Appendix, over \$57 million was generated in retail, food, and beverage sales in 2023. Identifying peaks during the holidays and summer would help with better staffing, maintenance, and retail planning.

2.37 million passengers by June 2026

To forecast monthly passenger volumes at Charlotte Douglas International Airport (CLT) for the upcoming year (June 2025 to May 2026), we implemented a Seasonal Auto Regressive Integrated Moving Average (SARIMA) model. This approach was chosen due to its strength in modeling both trends and seasonal patterns in time series data, which are particularly pronounced in airport passenger flows.

Using monthly passenger data from 2000 to 2024, we identified a consistent annual seasonality with peaks during summer and holiday months, and troughs in early spring and fall. The SARIMA model was fine-tuned through iterative testing of seasonal and non-seasonal parameters to ensure the best fit without overfitting. We selected the model with the lowest Akaike Information Criterion (AIC) score and validated it using residual diagnostics to confirm that the errors were randomly distributed.

The Seasonal ARIMA (SARIMA) model we selected reveals that monthly volumes are on track to exceed pre-pandemic levels by mid-2026 (2.37 million passengers by June 2026). This should help CLT with concession planning, infrastructure, staffing and investment decisions based on how busy the airport would be based on our projections.



STRATEGIC RECOMMENDATIONS

To ensure that CLT's leadership fully capitalizes on these insights, we recommend a multi-level approach based on operations and finance:

SCHEDULING & SHARING

Firstly, the delay prediction model should be integrated into the airport's scheduling systems and shared with airline partners (all of them preferably). This is especially important as on-time performance dips have direct financial implications (CLT estimates a \$2.5 million impact from delayed enplanements in peak 2023 months alone).

STAFFING

Second, the airport's staffing models should reflect predicted demand variations derived from the forecasting tool. Seasonal passenger volumes at CLT fluctuate by over 30%, yet fixed staffing levels persist across many operational areas. We can operate with a more flexible staffing regiment to capitalize on the dips. Predictive scheduling informed by monthly forecasts can lower overtime expenditure (currently it accounts for \$3.2 million annually).

COMMERCIAL VENDORS

Third, commercial vendors operating within CLT can benefit from earlier visibility into expected footfall. With retail revenue exceeding \$28 million and food and beverage sales surpassing \$29 million in 2023 (CLT Appendix), even a modest 5% increase in conversion driven by smarter promotion timing and layout optimization would yield over \$1.4 million in additional annual revenue.

INFRASTRUCTURE

Fourth, infrastructure planning must remain tightly coupled with the passenger forecasts. Delays are heavily concentrated at specific times of the day and aircraft types (narrow-body regional jets contributed disproportionately to taxi-out delays). Future apron redesigns or gate expansions should explicitly consider these patterns to maximize return on investment. Incorporating data on aircraft manufacturer, model generation, and wind sensitivity as explored in our model will enable more targeted long-term planning.

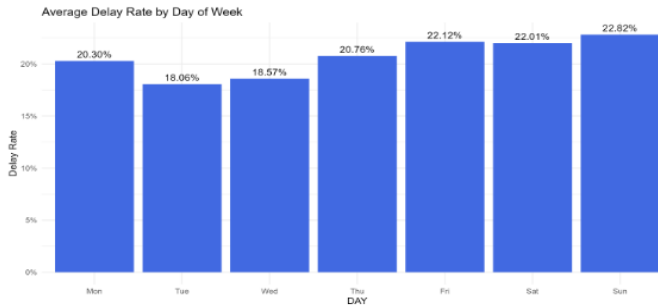
REGULATORY & INCENTIVE LEVEL CHANGES

Lastly, insights from both models strengthen CLT's negotiating position with airlines and regulatory bodies. By demonstrating empirical trends in flight behavior and passenger volumes, CLT can advocate performance-linked incentives and support funding allocations for capacity upgrades.

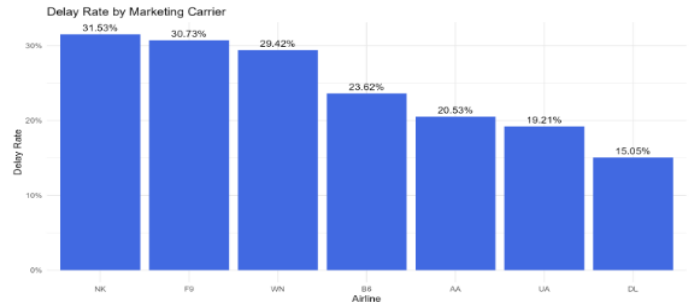


CLT Flight Delay Dashboard

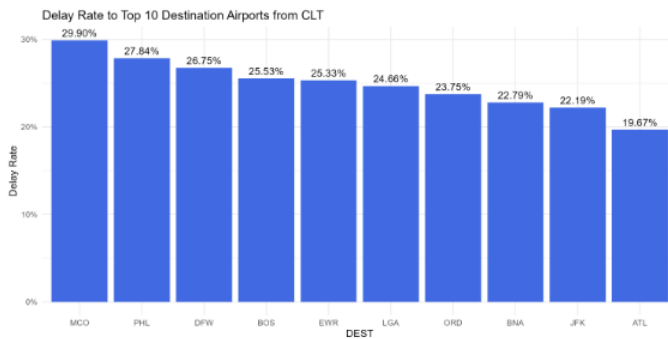
1. Delay Rate by Day of Week



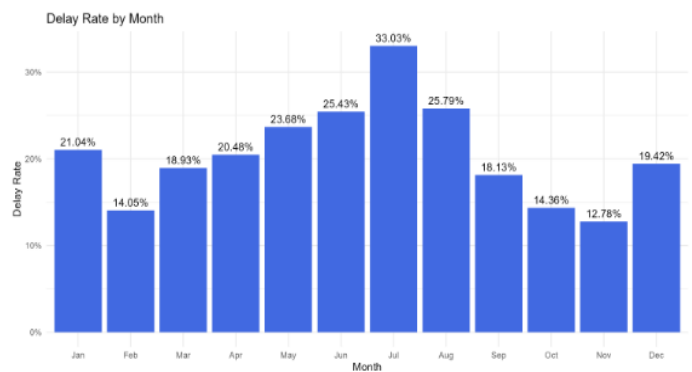
2. Delay Rate by Marketing Carrier



3. Delay Rate to Top 10 Destinations

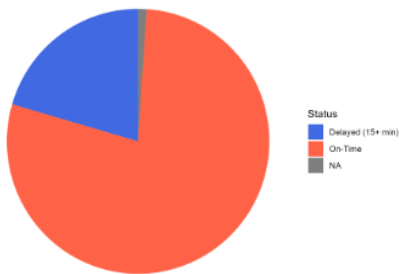


4. Delay Rate by Month



5. Flight Delay Distribution (2024)

Flight Delay Distribution (2024)



6. Delay based on Hour of Departure

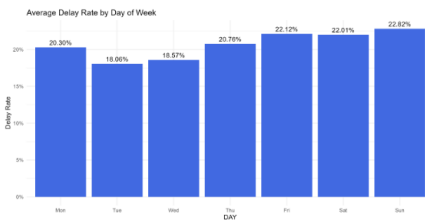


Dashboard: CLT 2024 Performance Overview: Showing American Airlines market dominance, peak delay routes, and missed retail opportunities. These insights guided our recommendations on vendor coordination and staffing optimization

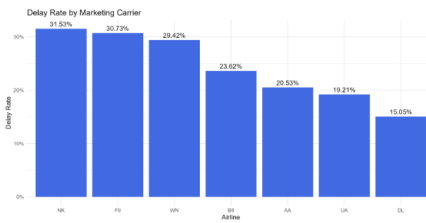
DASHBOARD VIEWED FROM SOURCE

CLT Flight Delay Dashboard

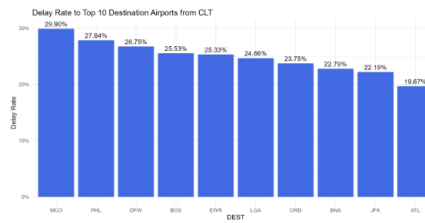
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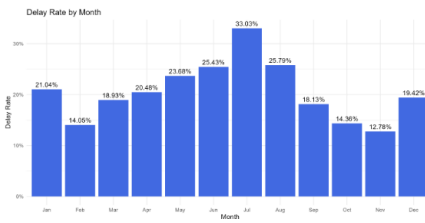
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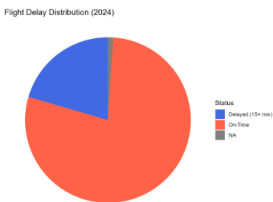
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6. Delay based on Hour of Departure



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